

Limiting distributions

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Definition

The period of a state i is the number

$$d_i = \gcd\{n \geq 1 : r_{ii}(n) > 0\}$$

If $d_i = 1$ we say that i is an aperiodic state.

- ▶ Observe that $r_{ii}(n) = 0$ if n is not a multiple of d_i .
(Conditional on $X_0 = i$, the probability of being in state i at time n is 0 if n is not a multiple of d_i .)
- ▶ d_i is the **greatest** integer with such property.
- ▶ Notice that if $p_{ii} > 0$, then $d_i = 1$.

Periodicity

Theorem

If $i \leftrightarrow j$ (the states i and j intercommunicate), then $d_i = d_j$.

- ▶ One can speak of the period of a whole class of states.
- ▶ For an irreducible Markov chain, we can refer to the period of that chain. If this period is 1 we say that the chain is **aperiodic**.

Periodicity

Proof:

First, the condition $i \leftrightarrow j$ means there exist integers k and m such that $r_{ij}(k) > 0$ and $r_{ji}(m) > 0$. Since $r_{ii}(k+m) \geq r_{ij}(k)r_{ji}(m) > 0$, d_i must divide $k+m$.

Now, let n be any integer with $r_{jj}(n) > 0$. Then $r_{ii}(k+n+m) \geq r_{ij}(k)r_{jj}(n)r_{ji}(m) > 0$, so d_i divides $k+n+m$. Since d_i also divides $k+m$, it follows that d_i divides n .

Thus, d_i divides any n for which $r_{jj}(n) > 0$, which implies that $d_j \geq d_i$. By interchanging i and j , it follows that $d_i \geq d_j$.

Therefore, $d_i = d_j$.

Structure of a periodic chain

If an irreducible chain has period $d > 1$, the transitions between states show a cyclic pattern.

Theorem

An irreducible Markov chain, with state space S , has period $d > 1$ if and only if there exist d subsets $S_j \subseteq S$, $0 \leq j \leq d - 1$, such that:

- ▶ $S_i \cap S_j = \emptyset$ if $i \neq j$ and $S = \bigcup_{j=0}^{d-1} S_j$.
- ▶ If $k \in S_j$ and $p_{kt} > 0$, then $t \in S_{j+1}$, where the subindices are taken modulo d .
- ▶ d is the largest integer for which such a decomposition of S exists.

Aperiodic chains

Theorem

For any finite and irreducible Markov chain, the following statements are equivalent.

- ▶ *The chain is aperiodic.*
- ▶ *The chain cannot be decomposed into a periodic structure as described in the previous theorem.*
- ▶ *There exists an integer $l \geq 1$ and a state j such that $r_{ij}(l) > 0$ for all $i \in S$.*
- ▶ *There exists an integer $m \geq 1$ such that $r_{ij}(m) > 0$ for all $i, j \in S$.*
- ▶ *For all $i \in S$, the greatest common divisor of the integers n for which $r_{ii}(n) > 0$ is 1.*

Aperiodic chains

- ▶ The existence of an integer $l \geq 1$ and a state j such that $r_{ij}(l) > 0$ for all $i \in S$ indicates that every element in the j -th column of P^l is strictly positive.

This result implies that state j is accessible from any $i \in S$ in exactly l steps.

- ▶ Similarly, if there exists an integer $m \geq 1$ such that $r_{ij}(m) > 0$ for all $i, j \in S$, then every entry of P^m is strictly positive. Consequently, this property holds for all $m' > m$ as well.

This result implies that for all $m' \geq m$, any state j is accessible from any state i in exactly m' steps.

Ergodic theorem

Theorem

If the Markov chain is irreducible and aperiodic, then for all $i, j \in S$,

$$\lim_{n \rightarrow \infty} r_{ij}(n) = \frac{1}{\mu_j}, \quad \text{independently of } i.$$

Therefore,

- ▶ *If the states are non-null recurrent and $(\pi_j = 1/\mu_j : j \in S)$ is the stationary distribution, then*

$$\lim_{n \rightarrow \infty} r_{ij}(n) = \pi_j, \quad \text{independently of } i.$$

- ▶ *If the states are transient or null recurrent, then*

$$\lim_{n \rightarrow \infty} r_{ij}(n) = 0$$

Ergodic theorem

Theorem

If the Markov chain is irreducible and has period $d > 1$, then the chain defined as

$$Y_n = X_{nd}, \quad n \geq 0$$

is aperiodic. Furthermore,

$$r_{jj}(nd) = \mathbb{P}(Y_n = j \mid Y_0 = j) \rightarrow \frac{d}{\mu_j} \quad \text{as } n \rightarrow \infty.$$

Doeblin's proof

We now present the proof of the ergodic theorem for the non-null recurrent case. Specifically, for any initial state i , it is necessary to demonstrate that

$$\lim_{n \rightarrow \infty} r_{ij}(n) = \pi_j$$

Doeblin's proof:

Let X be an irreducible, aperiodic, and non-null recurrent chain, and let Y be another Markov chain, independent of X , with the same transition probabilities.

Consider the composite chain $Z = (X, Y)$ defined by

$$Z_n = (X_n, Y_n), \quad n \geq 0$$

Doebelin's proof

The following properties can be established:

- ▶ Z is an irreducible Markov chain with transition probabilities

$$\mathbb{P}(Z_{n+1} = (j, \beta) \mid Z_n = (i, \alpha)) = p_{ij} p_{\alpha\beta}$$

- ▶ Z is non-null recurrent and aperiodic.
- ▶ The stationary distribution of Z is given by $\pi_{(j,k)} = \pi_j \pi_k$.

Consider an state s , and let $T_s = \min\{n \geq 1: Z_n = (s, s)\}$.

Moreover, let $T = \min\{n \geq 1: X_n = Y_n\}$.

Since $T \leq T_s$ and Z is non-null recurrent, we have that

$\mathbb{P}(T < \infty) = 1$. Hence, the random variable T is non-defective, and so

$$\mathbb{P}(T > n) \longrightarrow 0 \quad \text{as } n \longrightarrow \infty$$

Doebelin's proof

By the strong Markov property, we have that

$$\mathbb{P}(X_n = j \mid T \leq n) = \mathbb{P}(Y_n = j \mid T \leq n)$$

Therefore, $\mathbb{P}(X_n = j, T \leq n) = \mathbb{P}(Y_n = j, T \leq n)$.

Then

$$\begin{aligned}\mathbb{P}(X_n = j) &= \mathbb{P}(X_n = j, T \leq n) + \mathbb{P}(X_n = j, T > n) \\ &= \mathbb{P}(Y_n = j, T \leq n) + \mathbb{P}(X_n = j, T > n) \\ &\leq \mathbb{P}(Y_n = j) + \mathbb{P}(T > n)\end{aligned}$$

Analogously, $\mathbb{P}(Y_n = j) \leq \mathbb{P}(X_n = j) + \mathbb{P}(T > n)$.

Doebelin's proof

Therefore, as $n \rightarrow \infty$,

$$|\mathbb{P}(X_n = j) - \mathbb{P}(Y_n = j)| \leq \mathbb{P}(T > n) \rightarrow 0$$

Now, let $X_0 = i$ and let Y_0 have the stationary distribution ($\pi_j = 1/\mu_j: j \in S$). Then, since

$$\mathbb{P}(X_n = j) = r_{ij}(n), \quad \mathbb{P}(Y_n = j) = \pi_j$$

we conclude that $|r_{ij}(n) - \pi_j| \rightarrow 0$, as $n \rightarrow \infty$. Equivalently,

$$r_{ij}(n) \rightarrow \pi_j, \quad \text{independently of } i$$

Doeblin's proof

From $r_{ij}(n) \rightarrow \pi_j$ we can also conclude the following fact:

Independently of the initial distribution,

$$\begin{aligned}\lim_{n \rightarrow \infty} \mathbb{P}(X_n = j) &= \lim_{n \rightarrow \infty} \sum_{i \in S} r_{ij}(n) \mathbb{P}(X_0 = i) \\ &= \sum_{i \in S} \left(\lim_{n \rightarrow \infty} r_{ij}(n) \right) \mathbb{P}(X_0 = i) \\ &= \sum_{i \in S} \pi_j \mathbb{P}(X_0 = i) \\ &= \pi_j \sum_{i \in S} \mathbb{P}(X_0 = i) = \pi_j\end{aligned}$$

Limiting and stationary distribution

Theorem

If the Markov chain is finite, and for all $i, j \in S$,

$$\lim_{n \rightarrow \infty} r_{ij}(n) = \pi_j, \text{ independently of } i,$$

then $(\pi_j: j \in S)$ is a stationary distribution and it is unique.

Proof:

First, notice that $\pi_j \geq 0$ because $r_{ij}(n) \geq 0$. Let $n \rightarrow \infty$ in the Chapman-Kolmogorov equations $r_{ij}(n+1) = \sum_{k \in S} r_{ik}(n) p_{kj}$ to get

$$\pi_j = \sum_{k \in S} \pi_k p_{kj}$$

Limiting and stationary distribution

Moreover, since $\sum_{j \in S} r_{ij}(n) = 1$ we have that $\sum_{j \in S} \pi_j = 1$ (again, let $n \rightarrow \infty$).

Hence, $(\pi_j: j \in S)$ is a stationary distribution.

Finally, to prove uniqueness, suppose that $(u_j: j \in S)$ is also a stationary distribution. Thus $u_j = \sum_{k \in S} u_k p_{kj}$ and, more generally,

$$u_j = \sum_{k \in S} u_k r_{kj}(n), \quad n \geq 1$$

Letting $n \rightarrow \infty$, we obtain

$$u_j = \sum_{k \in S} u_k \pi_j = \pi_j \sum_{k \in S} u_k = \pi_j$$

Limiting and stationary distribution

For a general Markov chain, the previous theorem is as follows:

Theorem

If for all $i, j \in S$ we have that

$$\lim_{n \rightarrow \infty} r_{ij}(n) = \pi_j, \text{ independently of } i,$$

then

- ▶ $\sum_{k \in S} \pi_k \leq 1$ and $\pi_j = \sum_{k \in S} \pi_k p_{kj}$ for all $j \in S$.
- ▶ Either $\pi_j = 0$ for all $j \in S$, or else $\sum_{j \in S} \pi_j = 1$.
- ▶ If all $\pi_j = 0$, there is no stationary distribution.

If $\sum_{j \in S} \pi_j = 1$, then $(\pi_j : j \in S)$ is the unique stationary distribution.

A birth and death process

Example: Consider a Markov chain such that $S = \mathbb{N}$ and

$$p_{ij} = \begin{cases} 1 - b, & j = i = 0 \\ b, & j = i + 1, i \geq 0 \\ d, & j = i - 1, i \geq 1 \\ 1 - b - d, & j = i, i \geq 1 \end{cases}$$

(We assume $b > 0$, $d > 0$, and $b + d \leq 1$.)

- ▶ The chain is irreducible and aperiodic because $p_{00} > 0$.
- ▶ Let $\rho = b/d$. The detailed balance equations imply:

$$\pi_{j+1} = \rho \pi_j, \quad j \geq 0$$

We deduce that for all $j \geq 0$, $\pi_j = \pi_0 \rho^j$.

A birth and death process

► Case $\rho < 1$.

Condition $\sum_{j \in S} \pi_j = 1$ implies

$$1 = \pi_0 \sum_{j=0}^{\infty} \rho^j = \frac{\pi_0}{1 - \rho}$$

Therefore, $\pi_0 = 1 - \rho$ and, hence,

$$\pi_j = (1 - \rho) \rho^j, \quad j \geq 0$$

There exists a stationary distribution and the chain is **non-null recurrent**. Furthermore,

$$\lim_{n \rightarrow \infty} r_{ij}(n) = (1 - \rho) \rho^j, \text{ independently of } i$$

A birth and death process

► Case $\rho \geq 1$.

If $\pi_0 = 0$, then $\pi_j = 0$ for all $j \geq 0$.

Conversely, if $\pi_0 > 0$, then $\pi_0 \sum_{j=0}^{\infty} \rho^j = \infty$.

In any case, if $\rho \geq 1$ there is no stationary distribution and all the states are either transient or null recurrent. Moreover,

$$\lim_{n \rightarrow \infty} r_{ij}(n) = 0, \quad i, j \geq 0$$

It can be proved that for $\rho = 1$ the chain is **null recurrent**, whereas for $\rho > 1$ it is **transient**.